Hence we have

$$\left(\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}+\varepsilon\mathcal{L}_{H}^{\dagger}\right)\left(\rho_{\beta}(f_{2,0,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})+\varepsilon f_{2,1,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})+\ldots)\right)=-\varepsilon w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta},$$

and equating powers of the reciprocal friction ε gives us a sequence of equations to solve

$$\begin{aligned}
&\text{Order 1:} && \bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,0,[\![\mathrm{OABAO}]\!]}\right)=0,\\
&\text{Order } \varepsilon\text{:} && \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,0,[\![\mathrm{OABAO}]\!]}\right)+\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,1,[\![\mathrm{OABAO}]\!]}\right)=-w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta},\\
&\text{Order } \varepsilon^{2}\text{:} && \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,1,[\![\mathrm{OABAO}]\!]}\right)+\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,2,[\![\mathrm{OABAO}]\!]}\right)=0,\\
&\text{Order } \varepsilon^{3}\text{:} && \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,2,[\![\mathrm{OABAO}]\!]}\right)+\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,3,[\![\mathrm{OABAO}]\!]}\right)=0,\\
&&&\vdots
\end{aligned}$$

The first equation at order one tells us that $f_{2,0,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})$ must be a function of position only, as the kernel of $\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}$ is the set of functions that are constant with respect to p. Hence we shall denote

$$f_{2,0,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})=\nu(\boldsymbol{q}),$$

where we our goal is now to explicitly find $\nu(\boldsymbol{q})$. Unfortunately, $\nu(\boldsymbol{q})$ is not explicitly determined at this point, we only know that it is a function of position only such that $\langle\nu\rangle=0$. This function will be determined as we solve the higher order equations.

In order to determine $\nu(\boldsymbol{q})$, we will have to solve the next PDE at order ε,

$$\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,1,[\![\mathrm{OABAO}]\!]}\right)+\mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,0,[\![\mathrm{OABAO}]\!]}\right)=-w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta},\tag{7.32}$$

where we will make use of the formulas

$$\begin{aligned}
\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}\nabla^{3}U:\boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)&=-3\rho_{\beta}\nabla^{3}U:\boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}+\frac{6\rho_{\beta}}{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U,\\
\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U\right)&=-\rho_{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U,\\
\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U\right)&=-\rho_{\beta}\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U,
\end{aligned}$$

to write

$$w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta}=\frac{\beta}{24}\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\left[6\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U-\frac{1}{3}\nabla^{3}U:\boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}-\frac{2}{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U\right]\rho_{\beta}\right).\tag{7.33}$$

Additionally, for $\nu(\boldsymbol{q})$ we have the identity

$$\mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}\nu\right)=-\boldsymbol{p}^{T}\nabla\nu\rho_{\beta}=\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\left[\boldsymbol{p}^{T}\nabla\nu+\mu(\boldsymbol{q})\right]\rho_{\beta}\right),\tag{7.34}$$